

Computational Work 2: Cholesky Factorization

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Course: Computational Matrix Methods

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Abstract

This work investigates two computational problems involving numerical linear algebra and the numerical solution of a one-dimensional steady-state heat conduction problem. First, a symmetric matrix generated according to the prescribed procedure is analyzed to determine whether it is positive definite, followed by the computation of its Cholesky factorization, condition number, and the solution of the associated linear system. Second, the steady-state heat conduction equation in a unit-length bar is discretized using the finite difference formulation specified in the problem statement, resulting in a linear system whose solution provides the temperature distribution along the bar. The numerical results are analyzed with respect to the algebraic properties of the matrices, the conditioning of the linear systems, and the physical interpretation of the computed temperature profile. All problem-specific data are generated deterministically from the student's enrollment number according to the procedures provided in the assignment.

1 Problem

The computational work is divided into two problems involving numerical linear algebra and the numerical solution of a steady-state heat conduction problem.

1.1 Problem 1: Symmetric Positive Definite Matrix

Let $A \in \mathbb{R}^{10 \times 10}$ and $b \in \mathbb{R}^{10}$ be generated according to the procedure given in Code 1 of Appendix A.1, using the student's enrollment number as the random seed.

The following tasks are considered:

1. Determine whether the matrix A is symmetric positive definite. If so, compute its Cholesky factorization

$$A = GG^T,$$

and present the Cholesky factor G .

2. Compute the condition number $\kappa(A)$ and justify the result.
3. Solve the linear system

$$Ax = b.$$

1.2 Problem 2: One-Dimensional Steady-State Heat Conduction

Consider the one-dimensional steady-state heat conduction problem in a unit-length bar,

$$-k \frac{d^2 T}{dx^2} = f, \quad 0 \leq x \leq 1, \quad (1)$$

where $T(x)$ denotes the temperature, $k > 0$ is the thermal conductivity, and f is a constant source term.

The boundary conditions are determined according to the procedure given in Code 2 of Appendix A.2. One of the following cases is selected:

$$\text{Case 1: } \left. \frac{dT}{dx} \right|_{x=0} = 0, \quad T(1) = 0,$$

$$\text{Case 2: } T(0) = 0, \quad T(1) = 0,$$

$$\text{Case 3: } \left. \frac{dT}{dx} \right|_{x=0} = 0, \quad \left. \frac{dT}{dx} \right|_{x=1} = 0.$$

Using the finite difference formulation specified in the assignment, with 11 grid points, the problem is represented by the linear system

$$MT = s, \quad (2)$$

where M is the matrix associated with the discretized differential operator, s is the source vector including the boundary-condition contributions, and T is the vector of unknown temperatures.

The objectives are:

1. Generate M and s using the student's enrollment number and solve the resulting linear system.
2. Present the numerical temperature distribution $T(x)$.
3. Plot the temperature distribution and interpret the result from a physical perspective.

2 Methodology

The computational experiments were performed using Python and the NumPy library. All problem-specific data were generated deterministically from the student's enrollment number, following the algorithms provided in the assignment.

2.1 Matrix Generation and Cholesky Factorization

For the first problem, the matrix $A \in \mathbb{R}^{10 \times 10}$ is generated according to

$$A = BB^T + 10I, \quad (3)$$

where B is a randomly generated integer matrix and I is the 10×10 identity matrix. The vector $b \in \mathbb{R}^{10}$ is independently generated using the prescribed random seed.

The symmetry and positive definiteness of A are verified before computing its Cholesky factorization. When A is symmetric positive definite, the factorization

$$A = GG^T \quad (4)$$

is computed, where G is a lower triangular matrix with positive diagonal entries.

The condition number of A is then computed to quantify the sensitivity of the corresponding linear system to perturbations. Finally, the system

$$Ax = b \quad (5)$$

is solved numerically.

2.2 Heat Conduction Problem

For the second problem, the parameters k , f , and the boundary-condition type are generated using the prescribed procedure and the enrollment number as the random seed. The spatial domain $[0, 1]$ is represented by 11 equally spaced grid points, resulting in a grid spacing

$$h = \frac{1}{10}. \quad (6)$$

The matrix M and vector s are constructed according to the finite difference formulation specified in the assignment. The resulting linear system

$$MT = s \quad (7)$$

is then solved numerically to obtain the temperature vector

$$T = [T_0 \quad T_1 \quad \cdots \quad T_{10}]^T. \quad (8)$$

The numerical temperature values are associated with the corresponding spatial grid points and plotted as a function of x . The resulting temperature distribution is subsequently interpreted in terms of the physical behavior of the one-dimensional heat conduction problem.

All numerical computations were performed using double-precision floating-point arithmetic.

A Generation Codes

The following codes reproduce the procedures provided in the assignment for generating the numerical data used in the computational experiments.

A.1 Code 1: Matrix and Vector Generation

Listing 1: Generation of the matrix A and vector b using the student's enrollment number.

```
import numpy as np

def gerar_matriz(matricula , n=10):
    np.random.seed(matricula)
    B = np.random.randint(-5, 6, size=(n, n))
    A = B @ B.T + n * np.eye(n)
    return A

def gerar_vetor(matricula , n=10):
    np.random.seed(matricula + 1)
    return np.random.randint(-20, 21, size=n)

matricula = 31837 # student's enrollment number

A = gerar_matriz(matricula)
b = gerar_vetor(matricula)

print("Matriz A:")
print(A)

print("\nVetor b:")
print(b)
```

A.2 Code 2: Heat Conduction Problem

Listing 2: Generation of the finite difference system for the one-dimensional heat conduction problem.

```
import numpy as np

def gerar_matriz_por_matricula(matricula , n=11):
    np.random.seed(matricula)

    h = 1 / (n - 1)
    M = np.zeros((n, n))
```

```

k = np.random.uniform(0.05, 0.3)
fonte = np.random.uniform(0.5, 2.0)

tipo_bc = np.random.choice([
    "neumann-dirichlet",
    "dirichlet-dirichlet",
    "neumann-neumann"
])

coef = 1 / h**2
s = np.ones(n) * (h**2 / k) * fonte

for i in range(1, n - 1):
    M[i, i - 1] = coef
    M[i, i] = -2 * coef
    M[i, i + 1] = coef

if tipo_bc == "neumann-dirichlet":
    M[0, 0] = -coef
    M[0, 1] = coef
    M[-1, -1] = 1
    s[-1] = 0

elif tipo_bc == "dirichlet-dirichlet":
    M[0, 0] = 1
    M[-1, -1] = 1
    s[0] = 0
    s[-1] = 0

elif tipo_bc == "neumann-neumann":
    M[0, 0] = -coef
    M[0, 1] = coef
    M[-1, -2] = -coef
    M[-1, -1] = coef

return M, s, k, fonte, tipo_bc

```

matricula = 31837 # student's enrollment number

```
M, s, k, fonte, bc = gerar_matriz_por_matricula(matricula)
```

```
print("Matriz_M:\n", M)
```

```
print("Vetor_s:\n", s)
```

```
print("k=", k)
```

```
print("fonte=", fonte)
```

```
print("BC=", bc)
```